

Experimental design and analysis on Dice Superstitions

Math 261B Project

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EXECUTIVE SUMMARY

Superstitious beliefs are apparent in many aspects of human behavior to attempt to favorably change the outcome of a particular event. For example, a sports fan may refuse to change what clothes they were wearing when their favorite team won a big game. We were interested in investigating if these sorts of behaviors have any basis in reality, or if they're a complete waste of time and energy. The experiment outlined in this report is designed to analyze if different superstitious behaviors applied to rolling a pair of dice affect the outcome of the dice roll.

We researched what common superstitious behaviors are when it comes to rolling dice, and we decided to test what seem to be the three most common. The behaviors we tested include rubbing the dice before rolling, blowing on the dice before rolling, and shouting the desired outcome before rolling, as well as every combination of these three behaviors in conjunction with each other. We also tested not doing anything before rolling the dice to serve as a control group to compare to.

We applied each combination of superstitions to three different pairs of dice 50 times each, for a total of 1200 rolls. For each of the 1200 iterations, the summation of the dice roll was observed and recorded. Then, statistical procedures were used to test if there were any significant differences between any of the superstitious techniques used or not doing anything at all on the outcome of the dice roll. We believe that if there is any merit behind superstition in general, we would be able to perceive it in our tests, as we chose three very common behaviors one might see when people roll dice in real life.

The benefit of our experimental design is that it was relatively fast and easy to collect data. If further investigation is needed, we could easily collect more data in a similar manner. A potential draw-back is that it would be impossible for us to test every superstition in every possible circumstance.

The results of the statistical analysis showed a lack of evidence to support the claim that any of the superstitious behaviors we tested impact a dice roll. Our recommendation is to not concern yourself with such behaviors when it comes to events you can't control. Perhaps it is human nature to want to be in control. However, we found no evidence to suggest that that is actually possible in every situation. Examples of situations it is impossible to be in control over might include rolling dice, the outcome of a sporting event, whether the airplane you are traveling in will land or not, etc. A detailed overview of our experiment and our findings will follow in the full report.

PRE-EXPERIMENT

Statement of the problem

There exists a common belief among dice gamers that a die can be trained to bring fortune by several superstitions. The experiment of interest was chosen to test if superstitious behaviors has any effect on high or low rolls coming.

Null hypothesis: Superstitions have no effect on the outcome of dice rolls.

Alternative hypothesis: Superstitions have significant positive effects on the outcome of dice rolls.

Selection of response variable, factors and levels

We choose the total of scores achieved after rolling each pair of dice to be the response variable.

The observational units are the dice.

Factors	Levels		
	Pair 1	Pair 2	Pair 3
Dice (Blocking)			
Rub	Yes	No	
Blow	Yes	No	
Shout	Yes	No	

Table 1. Factors and Levels



Figure 1. Three different pairs of dice

To test this, we implemented a factorial design with four factors.

1. *Dice*: We used three pairs of dice from three different board games. The dice were all standard six-sided plastic dice. Since the variability among dice is not of experimental interest, dice are considered to be blocking variable.

2. *Rub*: This factor has two levels; “yes” or “no.” Either we rubbed the dice between our hands before throwing them, or we didn’t.
3. *Blow*: Similarly, this factor has two levels; “yes” or “no.” Either we blew on the dice before rolling them or we didn’t.
4. *Shout*: This factor has two levels; “yes” or “no” which implies either we shouted “12!” before rolling the dice or we didn’t.

Furthermore, we only randomly chose three pairs of dice out of millions of dice in the universe, therefore, *Dice* is random effect; *Rub*, *Blow* and *Shout* are fixed effects.

EXPERIMENT AND ANALYSIS

Choice of Experimental Design

We have one factor with 3 levels and two factors with 2 levels each, thus we have $3 \times 2 \times 2 \times 2 = 24$ “cells.” In each cell, we have 50 replicates for 1200 observations in total. Due to limited resources, we could not obtain 1200 pairs of dice to perform the experiment. Three-factor repeated measures design, therefore, was selected to tackle this issue.

Model used:

$$Y_{ijkm} = \mu + \tau_i + \alpha_j + \beta_k + \lambda_m + \alpha\beta_{jk} + \alpha\lambda_{jm} + \beta\lambda_{km} + \alpha\beta\lambda_{jkm} + \epsilon_{ijkm}$$

Assumptions

Prior to performing the experiment, we have normality and constant variance assumption on the errors.

Experimental Description

To minimize the variability between each roll, the whole experiment was conducted in the same environment and by one same individual.

The factor *Dice* is a random effect, because we chose 3 pair of dice which randomly happened to be in one of our homes out of every possible pair of dice in the world. It is also a blocking variable because we are not actually interested in testing if the different dice produce measurably different outcomes. Each pair of dice could be thought of as the “subjects” of our experiment, and they all received every possible combination of treatments.

The factors *Rub*, *Blow*, and *Shout* (the “treatments”) are all fixed effects. For each treatment, we tested every possible level of that treatment, either applying the superstition or not applying the superstition. The treatment *Shout* was not intended to test every possible phrase, one could shout as they roll a pair of dice. It is specifically designed to test shouting the outcome you

desire. In our experiment, we simply regarded higher rolls as more desirable than lower rolls. So, to test shouting, we chose to shout “12!”

Statistical Analysis

The following histograms show the distributions of the dice rolls when applying or not applying different treatments.

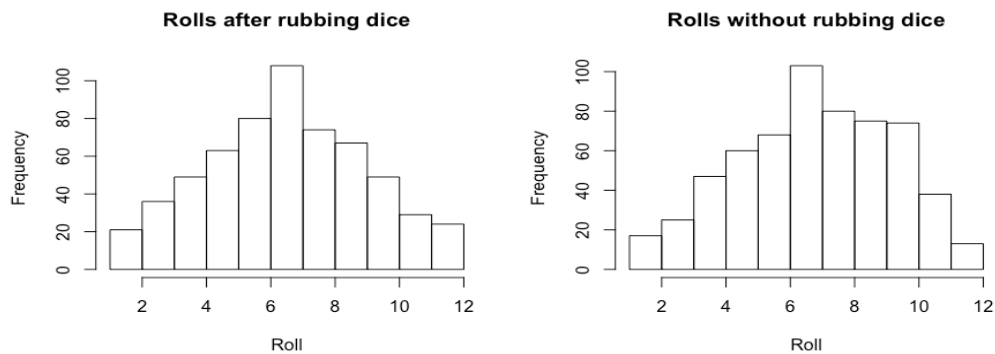


Figure 2. Histogram of dice with and without Rubbing

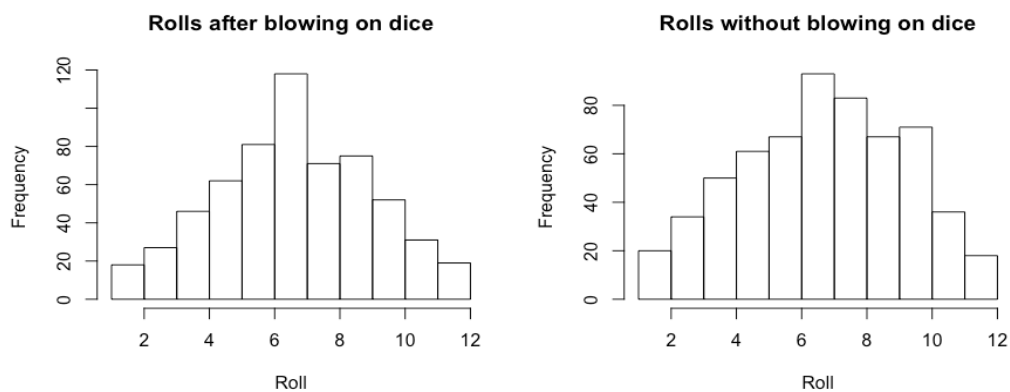


Figure 3. Histogram of dice with and without Blowing

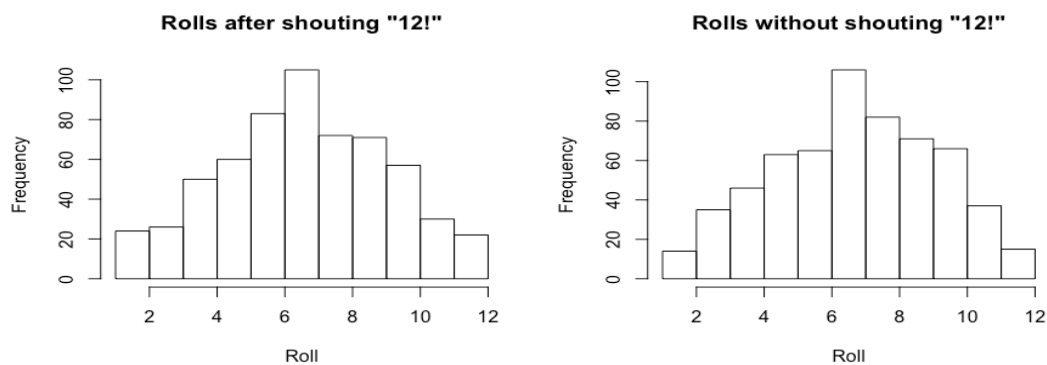


Figure 4. Histogram of dice with and without Blowing

Interestingly, it looks like the average roll was slightly *higher* in the trials where we didn't apply any superstitions. This is the exact opposite of what we would expect to see if the alternative hypothesis was true and superstition could positively affect the dice rolls. Perhaps superstition causes bad luck. Here is a cube plot of the mean of every combination of treatments.

In every case, *not* applying a treatment produced a higher mean roll.

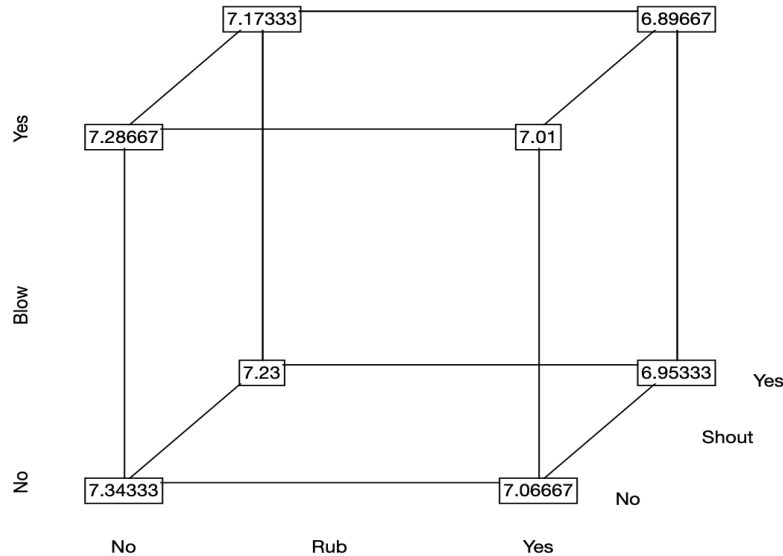


Figure 5. Geometric view of the mean of each treatment combinations

We ran analysis of variance to test if any of these differences were statistically significant.

Tests wrt Random Effects					
Source	SS	MS Num	DF Num	F Ratio	Prob > F
Rub	22.9633	22.9633	1	3.7988	0.0515
Blow	0.96333	0.96333	1	0.1594	0.6898
Shout	3.85333	3.85333	1	0.6375	0.4248
Rub*Blow	9.72	9.72	1	1.6080	0.2050
Rub*Shout	5.60333	5.60333	1	0.9270	0.3359
Blow*Shout	0.96333	0.96333	1	0.1594	0.6898
Rub*Blow*Shout	0.33333	0.33333	1	0.0551	0.8144
Dice&Random	0.915	0.4575	2	0.0757	0.9271

Figure 6. ANOVA for Three-factor Repeated Measures Design

Most of the p-values are large, indicating no significant effect, however the p-value for *Rub* was very close to 0.05. This raises a red flag and is something to further investigate.

According to statistician Ronald Fisher: “The test of significance only tells [us] what to ignore, namely all experiments in which significant results are not obtained...isolated significant results...are left in suspense pending further investigation.” (Fisher, 1929). In other words, if we see a small p-value, and we have the means to do so, we should perform another experiment to see if what we saw the first time was indeed rare or “unlucky,” or if it was common and therefore, we should reject our null hypothesis.

We collected more data (300 more rolls), this time only testing the effects of rubbing vs not rubbing, still with all three pairs of dice. We ran a 2-way ANOVA with *Dice* as a random blocking variable.

Tests wrt Random Effects					
Source	SS	MS Num	DF Num	F Ratio	Prob > F
Rub	3.85333	3.85333	1	0.6233	0.4304
Dice&Random	18.2867	9.14333	2	1.4790	0.2295

Figure 7. Two-way ANOVA of Randomized complete block design

Our second experiment suggests that rubbing the dice has no effect on the outcome of the rolls. Perhaps the data we collected in the first experiment was particularly rare to have observed. Given the evidence we have seen from both sets of data we collected, we feel we are unable to reject the null hypothesis. There is no evidence to support the claim that any of the superstitious behaviors we tested had any significant effects on the outcomes of the dice rolls.

We also conducted further investigation on the effect of shouting “12!” on the total scores yielded after each roll. Even though shouting “12!” at the dice didn’t seem to affect the average amount of each roll, we wanted to investigate if shouting “12!” produced more rolls of 12 than not shouting “12!” A contingency table was constructed as follows:

	Shout	Don’t shout	Total
12	22	15	37
Not 12	578	585	1163
Total	600	600	1200

Table 2. Contingency table of Shout and Roll generated 12

Shouting “12!” did actually produce more rolls of 12 than when we didn’t shout “12!” but we weren’t sure if it was statistically significant. We used the Chi-squared test of independence, which compares what we observed in each cell to what we might *expect* to see in each cell. Our null hypothesis was that shouting “12!” at the dice wouldn’t cause them to be more likely to produce a result of 12. The Chi-squared test returned a p-value of 0.3164, indicating a lack of significance. We failed to reject the null hypothesis and concluded that we had no evidence that shouting “12!” produced more rolls of 12.

Model Adequacy Checking

ANOVA models rely on the assumptions that the error terms of the model are normally distributed and have constant variance across the predicted values of our model. Also, they rely on the assumption that there is equal variance between each of the treatment groups.

The normal quantile plot in Figure 8 above shows the distribution of studentized residuals for our model. Since the studentized residuals roughly follow the theoretical quantiles of a normal distribution illustrated by the straight red line, the model seems to have error terms that are normally distributed. The residual by predicted plot above shows that the distribution of studentized residuals have a constant variance across predicted roll values in our model.

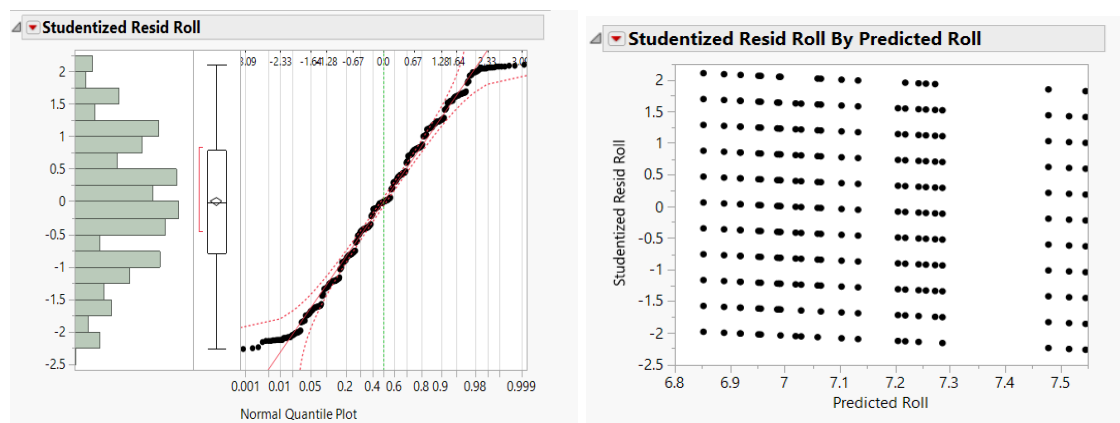


Figure 8. Normal Quantile Plot and Residual by Predicted Plot of Three-factor Repeated Measures Design

The residual plots in Figure 9 show the distribution of studentized residuals plotted by each of the treatments in our model. Judging by these plots, each treatment seems to have groups with equal variance. In summary, there is strong evidence that the assumptions of our models are satisfied.

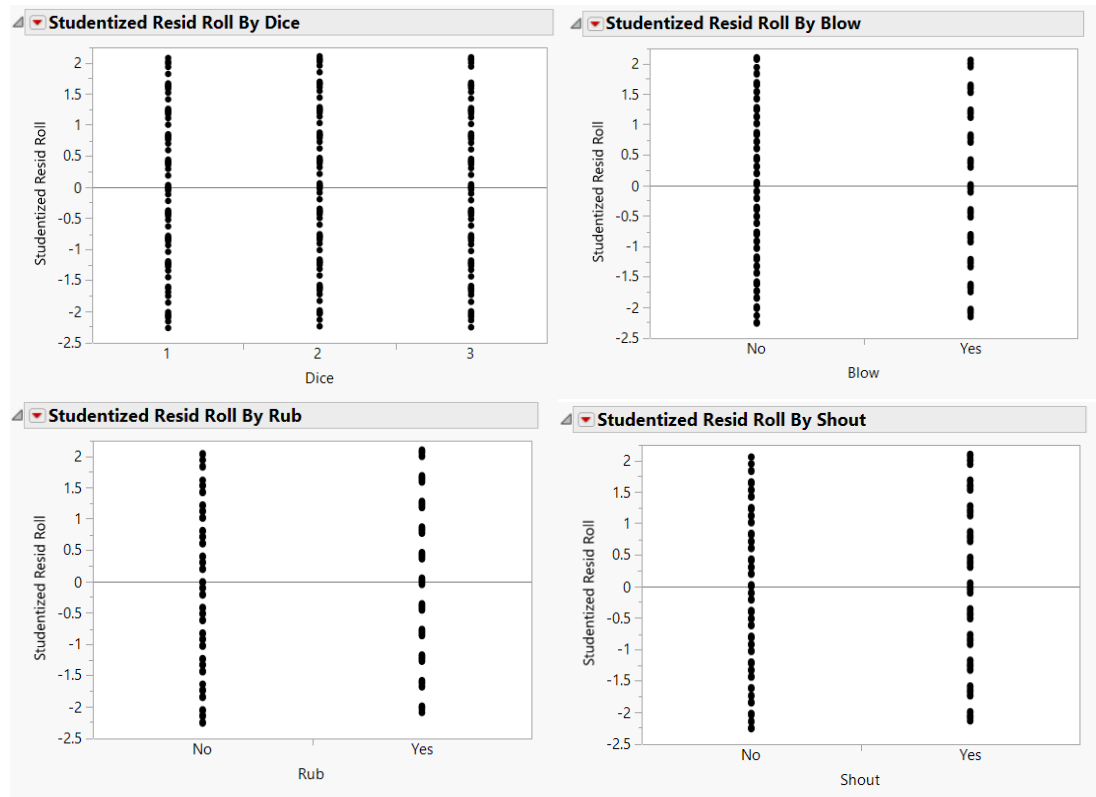


Figure 9. Studentized Residual Plot by Individual Treatments of Three-factor Repeated Measures Design

Conclusion

Our statistical analysis of the data indicates that the superstitious behaviors we tested have no statistically significant effect on the outcome of dice rolls. Our findings imply that the dice act completely independently of the person who is rolling them, and they behave in a fair and random manner.

Recommendation

Although none of the superstitions we tested turned out to have any significant effect, we are not able to infer anything about *all* superstitions, because the superstitions we chose were not chosen at random, and they don't have multiple randomly chosen levels. Much research remains to be done in this area. More large-scale and higher-budget experiments could be done testing more superstitions on the outcome of more events.

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